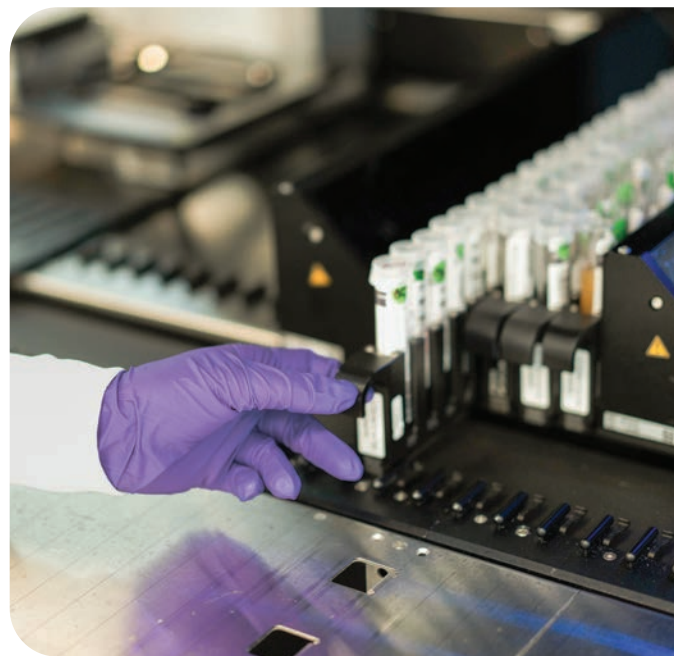




The new standard in prenatal care.

Fetal risk assessment for up to 14 recessive and X-linked genes, aneuploidies, and more — all from a single maternal blood draw.



The leader in cfDNA screening for recessive conditions.



Industry-Leading Performance

6 years

in clinical practice since the launch of UNITY Fetal Risk Screen in 2019

>600,000 tests

ordered¹

>95% sensitivity

for identifying affected fetuses²

99.8% NPV

negative predictive value³



Scientific Excellence

6 publications

in peer-reviewed journals

140,000+ pregnancies

clinically validated for a general obstetric population³

QCT™ technology

the only clinically validated cfDNA testing method for direct fetal risk assessment of recessive conditions⁴



Streamlined, Patient-Centric Workflow

1 test

single blood draw

2 panels

to suit your patient's needs

~2 weeks

from testing to results

Where others stop, we keep going.

UNITY Complete® is the **first-and-only clinically validated NIPT** to analyze cell-free DNA for a direct fetal risk assessment — including recessive conditions, aneuploidies, and more — all from a single maternal blood sample.^{3,5}

Partner testing is available, but not required to produce an accurate fetal risk assessment.

UNITY Complete®

Add-On

UNITY Fetal Risk™ Screen for up to 14 recessive and X-linked genes

- Cystic Fibrosisⁱ
- Spinal Muscular Atrophy
- Sickle Cell Disease
- Alpha-Thalassemia
- Beta-Thalassemia

*ACOG-aligned
screening
for a general
obstetric
population*

- Canavan Diseaseⁱ
- MCAD Deficiency
- Tay Sachs Diseaseⁱ
- Familial Dysautonomiaⁱ
- Smith-Lemli-Opitz Syndrome
- PMM2-Congenital Disorder of Glycosylation
- DMD-Associated Dystrophinopathiesⁱⁱ
- PAH Deficiency (Phenylketonuria)
- Fragile X Syndromeⁱⁱⁱ

UNITY Aneuploidy™ Screen for chromosomal conditions

- Trisomy21ⁱⁱⁱ
- Trisomy18ⁱⁱⁱ
- Trisomy13ⁱⁱⁱ
- Sex Chromosome Aneuploidies:
XO, XXY, XYY, XXX
- Zygoty *included for twins*
- 22q11.2 Microdeletion Syndromeⁱⁱⁱ
optional
- Fetal Sexⁱⁱⁱ *optional*

*Maternal carrier screening
with reflex to NIPT for
single gene conditions,
when indicated. Personalized
fetal risk provided for current
pregnancy.*

UNITY Fetal Antigen™ Tests for RhD & fetal antigen status

UNITY Fetal RhD NIPTⁱⁱⁱ for non-alloimmunized RhD-negative pregnancies

Provides fetal RhD status to guide medical management

UNITY Fetal Antigen NIPTⁱⁱⁱ for alloimmunized pregnancies

Provides fetal antigen status to guide medical management for hemolytic disease of fetus and newborn (HDFN) risk

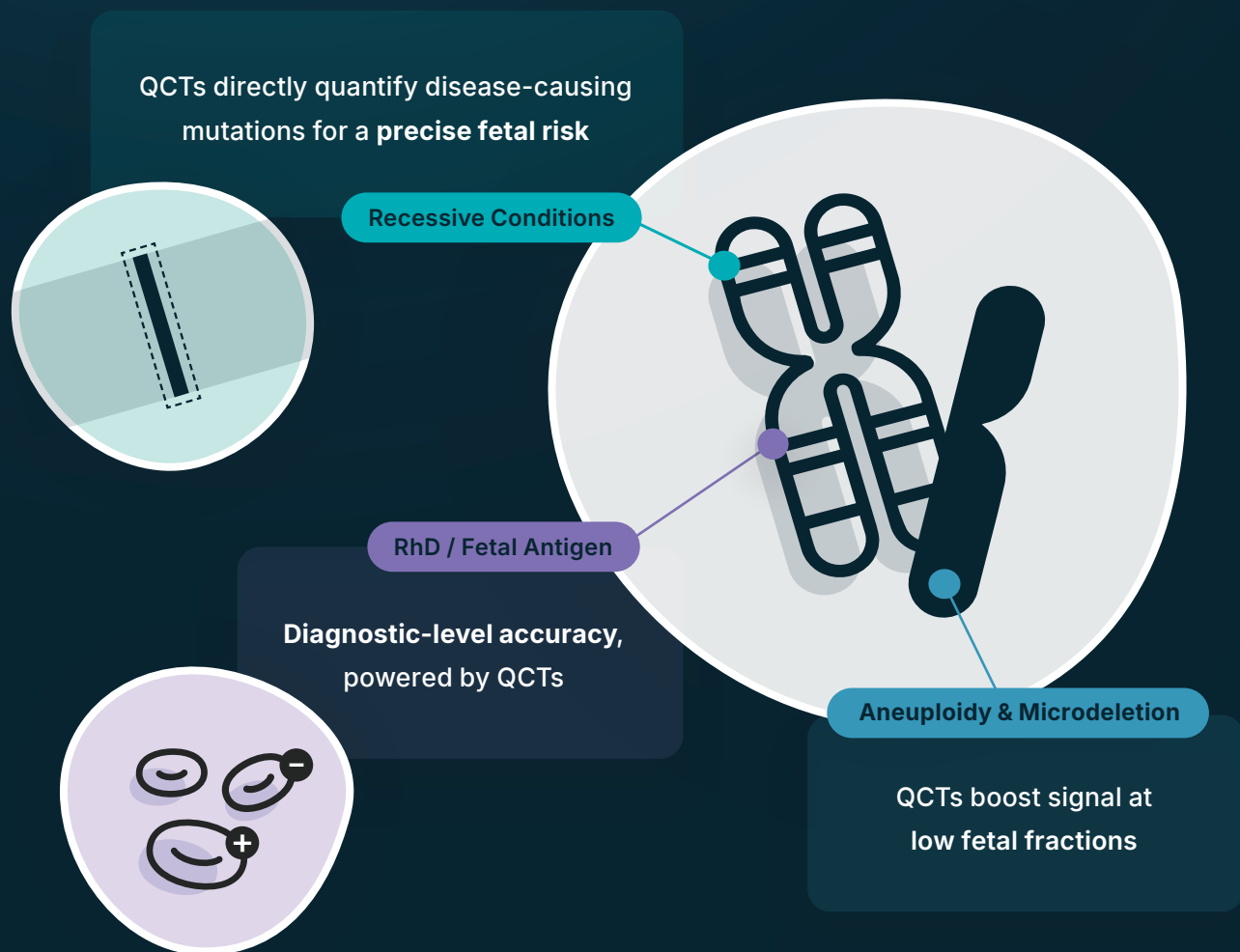
- big C
- little c
- D
- E
- Fy^a Duffy)
- K (Kell)

MCAD: Medium-Chain Acyl-CoA Dehydrogenase
PMM2: Phosphomannomutase 2
DMD: Duchenne Muscular Dystrophy
PAH: Phenylalanine Hydroxylase

i. ACOG recommended screening for Ashkenazi Jewish patients
ii. Fetal risk assessment via cfDNA is clarified via fetal sex only
iii. Available for mono and dizygotic twins

Direct insights to the fetus are possible with QCT™ technology.

Exclusive to UNITY, Quantitative Counting Template™ (QCT) technology does what other technologies cannot: precisely quantify the disease-causing mutation in cfDNA to deliver a direct fetal risk assessment for up to 14 recessive and X-linked genes with >95% sensitivity².



First-of-kind tests backed by clinical data.



TOP 10
clinical advancement
of 2023 by AJHG¹¹

nature
scientific reports



Analytical validity of single-gene NIPT with an estimated sensitivity of >98% and specificity of >99%⁵

OCT 2019

JME
Journal of Medical Economics



The cost to detect one affected pregnancy by UNITY Fetal Risk Screen was 62% lower than traditional carrier screening⁶

MAR 2022

American Journal of
Hematology



Accurately identified all sickle cell affected pregnancies as high risk at a greater than 9 in 10 risk⁷

APR 2022

**Genetics
in Medicine**



99.4% NPV and >90% sensitivity⁴

DEC 2022

PRENATAL
DIAGNOSIS



Assay sensitivity of 96% and NPV of 99.8%. 9-out-of-10 results were confirmed to be affected via neonatal outcomes³

SEP 2023

nature
scientific reports



Analytical sensitivity and specificity of >99.9%⁸

AUG 2023

**Annals of
Gynecology and
Obstetrics Research**



99.7% sensitivity & 99.9% specificity for autosomal trisomies. 80% of patients were <35 years⁹

MAY 2024

OBSTETRICS &
GYNECOLOGY



100% concordance with 465 neonatal outcomes¹⁰

JUL 2024

OBSTETRICS &
GYNECOLOGY



100% concordance with 401 clinical outcomes 5 *RHD*^ψ + 5 *RHD-CE-D* hybrid genes detected in non-alloimmunized RhD-neg patients¹¹

APR 2025

Journal of
**Cystic
Fibrosis**



Retrospective review of >100,000 consecutive general-risk pregnant patients. 100% sensitivity, >99.9% specificity¹²

SEP 2025

PREGNANCY



Critical titers were reached in ~70% of pregnancies regardless of fetal antigen (FA) status. Titers after cfDNA shows a FA-negative result may be misleading.

OCT 2025

Product

- UNITY Fetal Risk Screen
- UNITY Aneuploidy NIPT
- UNITY Fetal Antigen NIPT
- UNITY Fetal RhD NIPT

Publication Scope

- Analytical Validation
- Clinical Validation
- Health Economics Utility



Explore
our clinical
validation

UNITY Fetal Risk™ Screen

Simplified workflow. Superior Insights.

Designed with obstetric healthcare providers in mind.

Traditional Carrier Screening



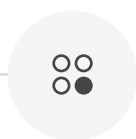
Maternal blood draw to determine carrier status



Post-test genetic counseling



Partner sample is required for reproductive risk
Less than 50% of partners complete carrier screening¹³



Couples receive a maximum **reproductive risk of 25%**



Post-test genetic counseling

UNITY Fetal Risk Screen



Maternal blood draw to determine carrier status **and fetal risk** for recessive conditions



Personalized fetal risk provided for this specific pregnancy (up to **9 in 10** or down to **1 in 10,000**)³



Post-test genetic counseling available with BillionToOne's licensed genetic counselors

While partner testing is accepted, it is not required for an accurate fetal risk assessment.

Trusted by providers nationwide with over 1 million UNITY tests ordered¹⁴.

Example Report

Dual panel flexibility in a single test

	Patient Carrier Status	cfDNA Fetal Risk Status
ACOG Guideline Panel		
Alpha-Thalassemia HBA1 HBA2	✔ Negative	N/A
Cystic Fibrosis CFTR	⊕ POSITIVE	Ⓛ Low Risk
Sickle Cell Disease / Beta-Thalassemia / Hemoglobinopathies HBB	✔ Negative	N/A
Spinal Muscular Atrophy SMN1	✔ Negative	N/A
Fragile X syndrome FMR1		
	✔ Negative 32 / 30 repeats	N/A
Plus Panel		
Canavan Disease ASPA	✔ Negative	N/A
DMD-Associated Dystrophinopathies DMD	✔ Negative	N/A
Familial Dysautonomia ELP1	✔ Negative	N/A
Medium-Chain Acyl-CoA Dehydrogenase Deficiency ACADM	✔ Negative	N/A
Phenylalanine Hydroxylase Deficiency PAH	✔ Negative	N/A
PMM2-Congenital Disorder of Glycosylation PMM2	✔ Negative	N/A
Smith-Lemli-Opitz Syndrome DHCR7	⊕ POSITIVE	Ⓜ HIGH RISK
Tay Sachs Disease HEXA	✔ Negative	N/A

Fetal risk **specific to this pregnancy** is assessed without a partner sample:

For positive carriers, fetal risks can be clarified down to **1 in 10,000...**

Risk Before cfDNA	Fetal Risk After cfDNA	
1 in 100 - 1 in 472	1 in 10,000	Low Risk

Ⓛ

99% of patients will receive a reassuring, low fetal risk result³

... or up to 9 in 10:

Risk Before cfDNA	Fetal Risk After cfDNA	
1 in 284 - 1 in 732	9 in 10	HIGH RISK

With easy-to-read results and fetal risk assessment, you can confidently counsel patients and support care decisions.



Early detection can make a difference.

Conditions Screened		Carrier Frequency ¹⁵	Available Interventions			
	Cystic Fibrosis	1 in 29-38				
	Spinal Muscular Atrophy	1 in 54				
	Sickle Cell Disease	1 in 8-33				
	Alpha-Thalassemia	1 in 112				
	Beta-Thalassemia	1 in 33				
	Canavan Disease	1 in 44-439				
	DMD-Associated Dystrophinopathies	1 in 717				
	Familial Dysautonomia	1 in 35-402				
	Medium-Chain Acyl-CoA Dehydrogenase Deficiency	1 in 67				
	PMM2-Congenital Disorder of Glycosylation	1 in 70				
	Phenylalanine Hydroxylase Deficiency (PKU)	1 in 79				
	Smith-Lemli-Opitz Syndrome	1 in 71				
	Tay Sachs Disease	1 in 28-193				
	Fragile X Syndrome	1 in 201				
	Gene or Enzyme Therapies Early detection enables access to therapies that may significantly improve outcomes		Dietary Modifications May improve outcomes and help alleviate symptoms		Multidisciplinary Care Early detection connects families with specialists for immediate postnatal care	 Early Intervention Programs Intervention programs and IEPs support development with demonstrated benefits

Risks shown are for general population. Ranges reflect carrier frequencies of specific populations with high prevalence of the condition (ex, Black, Ashkenazi Jewish).

Quantify fetal cfDNA for a precise fetal risk.

UNITY Fetal Risk Screen delivers highly accurate fetal risk assessment for inherited conditions in the general pregnant population and is able to identify affected pregnancies with both homozygous or heterozygous variants.

By both sequencing and directly quantifying disease-causing mutations in cfDNA from a single maternal blood sample, UNITY Fetal Risk Screen **detects approximately three times more affected pregnancies compared to traditional carrier screening.**

This efficient workflow removes barriers associated with partner testing and provides personalized fetal risk assessment early in pregnancy, making it an ideal solution for routine prenatal care across diverse patient populations.

Published in 2023
**PRENATAL
DIAGNOSIS**

Performance of single-gene noninvasive prenatal testing for autosomal recessive conditions in a general populations setting³

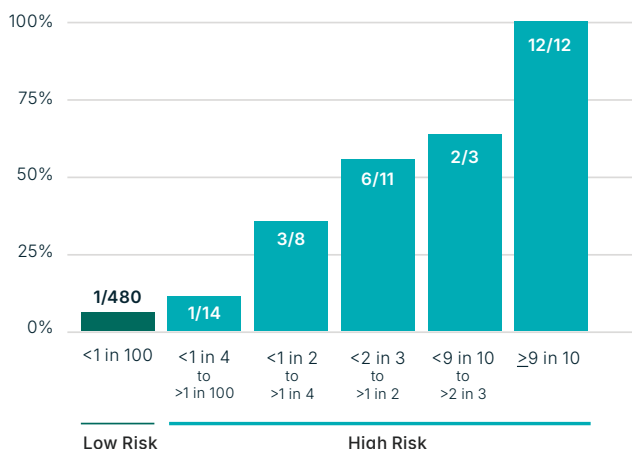
96.0% sensitivity Accuracy in detecting affected pregnancies with homozygous and heterozygous variants

99.8% NPV (negative predictive value) Trust in a negative result

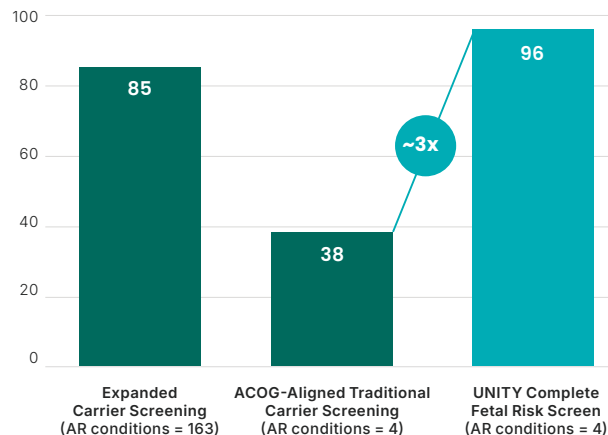
All cases identified as a 9 in 10 risk in the study cohort were confirmed to have an affected child.

UNITY overcomes challenges of traditional carrier screening, including lack of partner follow-up (58%) and misattributed paternity (~10%) — providing patients with timely and accurate fetal risk assessment.^{13, 16, 17}

Proportion of pregnancies confirmed to have an affected child



Number of affected fetuses per 100,000 pregnancies identified as high



UNITY Aneuploidy™ Screen

Optimized technology.
Designed for a general obstetric population.

- UNITY Aneuploidy Screen harnesses NGS (next generation sequencing), QCT, and SNP technology to deliver precise and highly accurate fetal risk assessment for chromosomal abnormalities.
- Aligned to current ACOG recommendations for aneuploidy screening
 - Standard panel includes T21, T18, T13, XO, XXY, XYY, and XXX
 - Zygosity included for all twin pregnancies to determine mono- versus dizygotic pregnancies and individual fetal sex

Published in 2024
**Annals of
Gynecology and
Obstetrics Research**

Performance Characteristics of a Next Generation Sequencing-Based cfDNA Assay for Common Aneuploidies in a General Risk Population⁹

**Mean Test
Characteristics**
n = 114,707

Maternal Age
29 years old

Gestational Age
13.9 weeks

Fetal Fraction
9.30% min 1.5%; max 39%

Average TAT
5 days

		Trisomy 21	Trisomy 18	Trisomy 13	Combined Autosomes
	Sensitivity	>99%	>99%	>99%	>99%
	Specificity	99.7%	>99.9%	>99.9%	99.9%
	PPV	>90%	>97%	>73%	>90%
	NPV	>99.9%	>99.9%	>99.9%	>99.9%

Expand your aneuploidy screening to accommodate diverse clinical needs with **flexible add-on options** that can be requested at any point during pregnancy, even post initial results.

Add-On

22q11.2 Microdeletion Syndrome

- Highly accurate detection through precise fetal cfDNA quantification
- Screens the full A-D region and certain nested microdeletions¹⁶
- >95% sensitivity, >99.9% specificity, and 80% positive predictive value (PPV)¹⁶

UNITY Fetal RhD & Antigen™ NIPT

200,000+ Fetal RhD and Antigen tests performed since 2020¹¹.

Add-On

UNITY Fetal RhD NIPT for non-alloimmunized RhD-negative pregnancies

Detects the presence or absence of the *RHD* gene deletion, common deletions, *RHD-CD-D* hybrid gene, and *RHDψ* via cfDNA^{8,10}

Traditional Workflow



Fetal RhD antigen status is **often unknown** without invasive procedure

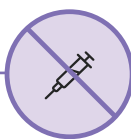


All RhD-negative mothers receive Rh₀(D) immune globulin

UNITY Fetal RhD NIPT



Determines fetal RhD antigen status



40% of fetuses are identified as RhD-negative; Rh₀(D) immune globulin not indicated¹⁸

Add-On

UNITY Fetal Antigen NIPT for alloimmunized pregnancies

Detects fetal antigen status for big C, little c, D, E, Fy^a (Duffy), and K (Kell) antigen(s)

**>99.9%
sensitivity¹⁰**

**>99.9%
specificity¹⁰**

**100%
concordance with
neonatal outcomes¹⁰**



Scan to
learn more

Your patients, our priority.

UNITY Complete streamlines testing through a single maternal blood draw, reducing the complexity of multiple clinic visits, partner testing, and numerous insurance bills and co-pays.

Patient-Centric Experience

- **Convenient self-service portal:** Track test progress and view results.
- **Educational support:** On-demand videos and resources.
- **Complimentary genetic counseling:** Virtual visits with our genetic counselors to review results.
- **Transparent, friendly billing practices:** Our US-based team contacts patients about balances before billing. One bill, one copay — no surprises.

Affordable for All

- **Wide insurance coverage:** In-network with the majority of insurance plans, covering over 200 million lives.
- **\$0 for all Medicaid patients:** 45% of UNITY patients have Medicaid¹⁹.
- **~80% of commercially insured patients pay less than \$200 for two tests (UNITY Fetal Risk Screen + Aneuploidy)**¹⁹.

Streamlined Provider Workflow

- **Single order:** One TRF for all UNITY Complete tests.
- **Portal or EHR integration:** Easy ordering and fast results via EHR or provider portal.
- **Actionable results:** Clear, reliable insights for confident clinical decisions.

1. Data on file as of May 2025. 2. For autosomal recessive conditions, the clinical sensitivity refers to the estimated clinical detection rate of high-risk fetuses by cfDNA fetal risk assessment. For X-linked conditions, for which cfDNA testing is only clarified via fetal sex, the clinical sensitivity refers to carrier screening detection and does not account for de novo mutations. 3. Wynn J, et al. Performance of single-gene noninvasive prenatal testing for autosomal recessive conditions in a general population setting. *Prenat Diagn.* 2023 Sep; 43(10):1344-1354. doi:10.1002/pd.6427. Epub 2023 Sep 6. PMID: 37674263. 4. Tsao, D. S., et al. (2019). A novel high-throughput molecular counting method with single base-pair resolution enables accurate single-gene NIPT. *Scientific Reports*, 9, 14382. https://doi.org/10.1038/s41598-019-50378-8. 5. Tsao, D. S., et al. (2019). A novel high-throughput molecular counting method with single base-pair resolution enables accurate single-gene NIPT. *Scientific Reports*, 9, 14382. https://doi.org/10.1038/s41598-019-50378-8. 6. Riku S. et al. (2022) Reflex single-gene non-invasive prenatal testing is associated with markedly better detection of fetuses affected with single-gene recessive disorders at lower cost, *Journal of Medical Economics*, 25:1, 403-411 DOI: 10.1080/13696998.2022.2053384. 7. Westin, E.R., et al. (2022), Validation of single-gene noninvasive prenatal testing for sickle cell disease, *Am J Hematol*, 97: E270-E273. doi:10.1002/ajh.26570. 8. Alford, Brian, et al. "Validation of a non-invasive prenatal test for fetal RhD, C, c, E, K and Fya antigens." *Scientific Reports* 13.1 (2023): 12786.. 9. Wynn J, et al. Performance Characteristics of a Next Generation Sequencing-Based cfDNA Assay for Common Aneuploidies in a General Risk Population. *Ann Gynecol Obstetr. Res.* 2024; 7(1): 102. doi:10.1002/ajh.26570. 10. Rego, Shannon, et al. "Cell-free DNA analysis for the determination of fetal red blood cell antigen genotype in individuals with alloimmunized pregnancies." *Obstetrics & Gynecology* (2022): 10-1097. 11. Mateus-Nino, Julio F., et al. "Clinical performance of cell-free DNA for fetal RhD detection in RhD-negative pregnant individuals in the United States." *Obstetrics & Gynecology* 145.4 (2025): 402-408. 12. Wynn, J., et al. "Routine cell-free DNA prenatal screening identifies pregnancies at high risk for cystic fibrosis that may benefit from fetal therapy." *Journal of Cystic Fibrosis* (2025). 13. Strauss, T S et al. "Barriers to completion of expanded carrier screening in an inner city population." *Genetics in medicine : official journal of the American College of Medical Genetics* vol. 25,7 (2023): 100858. doi:10.1016/j.jgm.2023.100858. 14. Internal data on file. Aug. 2025. 15. Carrier screening in the age of genomic medicine. Committee Opinion No. 690. American College of Obstetricians and Gynecologists. *Obstet Gynecol* 2017;129:e35-40. 16. UNITY Aneuploidy NIPT Validation. Data on File. January 2025. 17. Carlotti, K et al. "Perceived barriers to paternal expanded carrier screening following a positive maternal result: To screen or not to screen." *Journal of Genetic Counseling* Vol. 30,2 (2021): 470-477. doi:10.1002/jgc4.1333 Ghioffi CE, Goldberg JD, Haque IS, Lazarin GA, Wong KK. Clinical Utility of Expanded Carrier Screening: Reproductive Behaviors of At-Risk Couples. *J Genet Couns.* 2018 Jun;27(3):616-625. doi: 10.1007/s10897-017-0160-1. Epub 2017 Sep 27. PMID: 28956228; PMCID: PMC5943379. 18. Prevention of Rh D Alloimmunization. *Obstetrics & Gynecology* 130(2):p e57-e70, August 2017. DOI: 10.1097/AOG.0000000000002232. 19. BillionToOne, Inc. (2024). Internal billing data on file.



Contact Us

650.460.2551
support@unityscreen.com

UNITY Complete is a screening test, not a diagnostic test. Any high-risk result should be followed up with diagnostic testing such as CVS or amniocentesis.

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